1-Number of Zeros in a Given Array

Started on	Thursday, 8 October 2026, 10:18 PM
State	Finished
Completed on	Thursday, 8 October 2026, 10:20 PM
Time taken	1 min 58 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

uestion 1 | Correct Mark 1.00 out of 1.00 Flag question

roblem Statement

iven an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

nput Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

utput Format

First Line Contains Integer – Number of zeroes present in the given array.

nswer: (penalty regime: 0 %)

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n, count = 0;
    scanf("%d", &n);

    int a[n];

    for(int i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);

        if(a[i] == 0)
        {
            count++;
        }
    }

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔
✔	10 1 1 1 1 1 1	0	0	✔

2-Majority Element

Started on	Thursday, 8 October 2026, 10:20 PM
State	Finished
Completed on	Thursday, 8 October 2026, 10:23 PM
Time taken	2 mins 46 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

- n == nums.length
- 1 <= n <= 5 * 10^4
- -2^{31} <= nums[i] <= 2^{31} - 1

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int a[n];

    for(int i = 0; i < n; i++)
        scanf("%d", &a[i]);
    for(int i = 0; i < n; i++)
    {
        for(int j = i + 1; j < n; j++)
        {
            if(a[i] > a[j])
            {
                int temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }
    }
    printf("%d", a[n / 2]);
    return 0;
}
```

	Input	Expected	Got	
✔	3	3	3	✔

3-Finding Floor Value

Started on	Thursday, 8 October 2026, 10:24 PM
State	Finished
Completed on	Thursday, 8 October 2026, 10:25 PM
Time taken	1 min 20 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

uestion 1 | Correct Mark 1.00 out of 1.00 Flag question

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

nput Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n, x;
    scanf("%d", &n);

    int a[n];

    for(int i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    scanf("%d", &x);

    int floor = -1;

    for(int i = 0; i < n; i++)
    {
        if(a[i] <= x)
        {
            floor = a[i];
        }
        else
        {
            break;
        }
    }

    printf("%d", floor);

    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔

4-Two Elements sum to x

Started on	Thursday, 8 October 2026, 10:26 PM
State	Finished
Completed on	Thursday, 8 October 2026, 10:27 PM
Time taken	1 min 19 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

```c
#include <stdio.h>

int main()
{
    int n, x;
    scanf("%d", &n);

    int a[n];

    for(int i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    scanf("%d", &x);

    for(int i = 0; i < n; i++)
    {
        for(int j = i + 1; j < n; j++)
        {
            if(a[i] + a[j] == x)
            {
                printf("%d\n", a[i]);
                printf("%d", a[j]);
                return 0;
            }
        }
    }

    printf("No");

    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10	No	No	✔

5-Implementation of Quick Sort

Started on	Thursday, 8 October 2026, 10:27 PM
State	Finished
Completed on	Thursday, 8 October 2026, 10:29 PM
Time taken	1 min 44 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int a[n];

    for(int i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    for(int i = 0; i < n - 1; i++)
    {
        for(int j = 0; j < n - i - 1; j++)
        {
            if(a[j] > a[j + 1])
            {
                int temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;
            }
        }
    }

    for(int i = 0; i < n; i++)
    {
        printf("%d ", a[i]);
    }

    return 0;
}
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.